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Measurements on polarized light

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A setup with a polariser turning lightbulb light into polarised light and an adjustable polarisation filter is used to confirm Malus' law $I(\theta) = I_0 \cos^2(\theta)$. Furthermore, the birefringence of a calcite crystal is examined by splitting the differently polarised light into an ordinary and an extraordinary ray. Additionally, the polarisation properties of a quarter-wave plate made out of quartz are explored in the context of the Jones formalism. Laurent's half-shade polarimeter is utilized to investigate the optical activity of sucrose for five different wavelengths. The measured optical activity is then used to determine the sucrose concentration of a solution. The Faraday effect is observed by applying a magnetic field on an N-SF6 glass body and sending light of different wavelengths through it. The Verdet constant is determined for each wavelength and subsequently compared to the theoretical model.

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1 Introduction

Light is an electromagnetic wave consisting of oscillating electric and magnetic fields perpendicular to each other and to the direction of propagation. It travels at a finite speed and exhibits both wave-like and particle-like behaviour. In the visible range (approx. 400–750 nm), light is essential for human vision and various optical technologies. One of the key characteristics of light is polarisation, which describes the orientation of the electric field. Natural light sources, such as the sun, emit unpolarised light, where the electric field oscillates in random directions. It light can become polarised through reflection, scattering, or by passing through a polarising filter.

In this experiment, different states of polarisation are examined. Methods to generate and analyse polarised light are explored using polarisers, birefringent materials, and wave plates. Additionally, phenomena like optical activity and the Faraday effect are used to investigate how polarisation can be influenced by substances or external magnetic fields.

Understanding polarisation and its manipulation is fundamental in optics and has practical applications in areas such as imaging, communication, and material analysis.

2 Generation and detection of linearly polarised light

Maxwell's equations form the basis for electro-/ magnetodynamics. They can be used to derive a wave equation that describes the behaviour of light:

$$\nabla^2 \vec{E} - \mu\varepsilon \frac{\partial^2 \vec{E}}{\partial t^2} = 0 \tag{1}$$

$$\nabla^2 \vec{B} - \mu\varepsilon \frac{\partial^2 \vec{B}}{\partial t^2} = 0 \tag{2}$$

The solution is a plane wave, whereby the following considerations are limited to the electric field:

$$\vec{E}(\vec{r}, t) = \vec{E}_0 e^{i(\vec{k}\cdot\vec{r} - \omega t)} \tag{3}$$

Light is a transverse wave. A distinction is made between different polarisation states during oscillation. If the direction of oscillation does not change within a period, this is referred to as linearly polarised light, whereby the electric field takes on the following form (propagation is considered in the z-direction):

$$\vec{E}(z, t) = (E_{0x}\,\hat{e}_x \pm E_{0y}\,\hat{e}_y)\cos(kz - \omega t) \tag{4}$$

If the electric field vector passes through a circular helix in one period, this is referred to as circularly polarised light. Then the electric field takes the following form:

$$\vec{E}(z,t) = E_0 \left(\cos(kz - \omega t)\, \hat{e}_x \pm \sin(kz - \omega t)\, \hat{e}_y\right) \quad (5)$$

Natural light is unpolarised, i.e., all polarisation directions occur equally frequently.

In this part of the experiment, different methods for generating linearly polarised light are investigated.

2.1 Linear polarisation by reflection

When unpolarised light strikes the surface of a dielectric material such as glass at the Brewster angle, the reflected light becomes linearly polarised. This occurs because at this angle, the reflected and refracted beams are perpendicular to each other. The component of the electric field that lies in the plane of incidence (p-polarisation) induces dipole oscillations in the surface charges of the dielectric. These oscillating dipoles cannot re-radiate (i.e., reflect) light in the direction of the reflected beam when that direction is perpendicular to the transmitted beam — a fundamental result of dipole radiation patterns. As a result, no p-polarised light is reflected. Only the s-polarised component (perpendicular to the plane) reflects — and thus the reflected light is linearly polarised with the electric field perpendicular to the plane of incidence.

Direction of polarisation: The polarisation plane of the reflected light is perpendicular to the plane of incidence. This can be confirmed experimentally by rotating an analyser and observing a complete extinction at a certain orientation.

Condition for zero reflection: For the reflected beam to vanish, the incident light must be fully polarised in the plane of incidence (p-polarised). At the Brewster angle, this polarisation is completely transmitted, and no light is reflected.

2.2 Linear polarisation by birefringence

Unpolarised light from an incandescent lamp is directed onto a calcite (Iceland spar) crystal. The light is split into two rays: the ordinary and extraordinary rays, due to birefringence. Birefringence, also known as double refraction, is an optical property of a material in which the refractive index depends on the polarization and propagation direction of light due to anisotropy in the material's internal structure.

Visualisation of the two rays: The two rays can be made visible by placing a screen or paper behind the crystal and observing two distinct spots. Alternatively, by placing a polarising filter (analyser) after the crystal and rotating it, each beam can be selectively extinguished, confirming their polarisation directions.

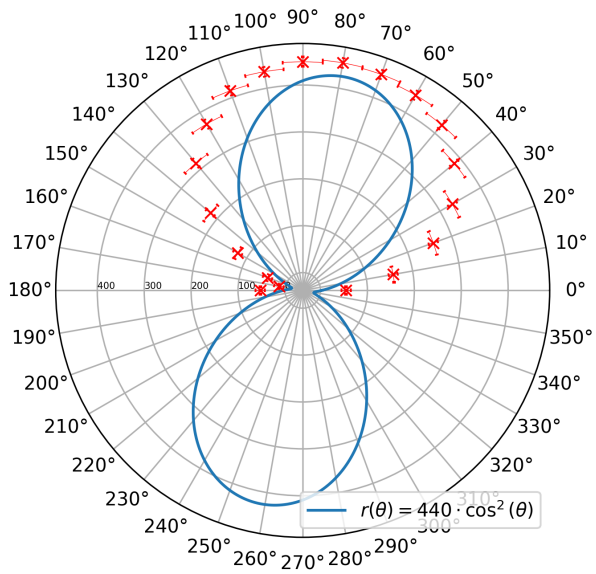

Figure 1: The graph shows our experimental values of the current and the theoretical trend predicted by Malus' law for light passing through a polariser. Since the current is proportional to the light intensity it should obey Eq. 6

Identification of the rays: The ordinary ray obeys Snell's law and remains fixed in direction relative to the crystal. The extraordinary ray deviates depending on the crystal orientation. By rotating the crystal and observing which ray shifts position, one can identify the extraordinary ray (the one that shifts) and the ordinary ray (the one that does not).

Polarisation state: The ordinary and extraordinary rays are linearly polarised perpendicular to each other. This can be confirmed using an analyser: as the analyser is rotated, each beam is extinguished at different analyser angles, confirming orthogonal linear polarisations.

Disappearance of birefringence: The birefringence effect disappears when the light propagates either parallel or perpendicular to the optical axis of the crystal. In the parallel case, the ordinary and extraordinary rays overlap and behave identically, and no splitting occurs. This can be tested by aligning the crystal such that its optical axis is either parallel or perpendicular to the incident beam and observing whether the double refraction persists.

3 Review of Malus' law

In this experiment, natural light from a light bulb shines on a polariser. A polariser has the property that it allows only one direction of polarisation to

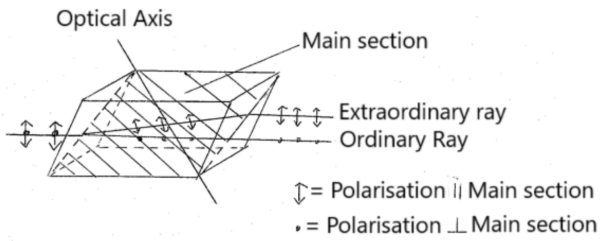

Figure 2: Birefringence on a calcite crystal.[2]

pass through. The polariser is placed such that the transmission axis is perpendicular to the propagation direction. Hence, the light behind the polariser is linearly polarised. A second polariser is placed behind the first polariser, whereby the transmission axis is rotated by an angle θ with respect to the transmission axis of the first polariser. The light also passes through the second polariser and the intensity of the light after the polariser is described by Malus' law:

$$I(\theta) = I_0 \cos^2(\theta) \tag{6}$$

where $I(\theta)$ is the intensity after the polariser and I_0 is the intensity of the electric field before the polariser. After the second polariser, the light hits a photocell, which results in an electric current that is measured with the multimeter Voltcraft VC 870. The current I is directly proportional to the light intensity $I(\theta)$.

In the experiment, the current is measured for different angles θ from 0° up to 180° in steps of 10°. A nonlinear fit, neglecting angle uncertainties, is performed using the following function to obtain parameters for the theoretical trend:

$$I(\theta) = I_0 \cos^2(\theta - \varphi) + I_{\min}$$

This nonlinear fit yields the parameters $I_0 = 422.5\text{A}$, $\varphi = 258.7°$ and $I_{\min} = 82.9\text{A}$. $I_{\min}$ is a current caused by scattered radiation and φ is an offset angle due to the mismatch between the zero angles of the linear polarisers. As shown in Fig.1, the measured intensity currents do not agree with Malus' law.
We do not know what could possibly have gone wrong, but after consulting our supervisor we should still present the experimental part in the protocol.

4 Birefringence in a calcite crystal and linear polarisation

In this experiment, natural (unpolarised) light from a light bulb is directed onto a calcite crystal and projected onto a screen. A distinction is made between *optically isotropic* and *optically anisotropic* media.

Optically isotropic media are characterised by the fact that the speed of light is independent of both the direction of propagation and the polarisation. In contrast, optically anisotropic media, such as calcite, exhibit *birefringence* — a phenomenon where light is split into two rays upon entering the material. The splitting is a consequence of material to posses a polarization- and propagation-dependent refractive index.

Calcite has a particular symmetry with respect to one axis, known as the optical axis. When a beam of light is incident perpendicularly on the surface of the calcite crystal such that the optical axis is tilted relative to the surface, the beam splits into two distinct rays: The ordinary ray (o-ray), which behaves according to Snell's law of refraction and has a constant refractive index. It is not influenced by the direction of the optical axis. The extraordinary ray (e-ray), which does not follow Snell's law and has a refractive index that depends on the angle between the propagation direction and the optical axis. The e-ray is refracted in a plane defined by the incident beam and the optical axis, known as the main section of the crystal.

After emerging from the calcite crystal, the ordinary and extraordinary rays are both linearly polarised, with their polarisation directions oriented at 90° to each other. This behaviour is illustrated in Fig.2.

If the light propagates either parallel or perpendicular to the optical axis inside the crystal, no beam splitting occurs: In the parallel case, the distinction between ordinary and extraordinary rays vanishes, and light behaves uniformly. In the perpendicular case, although no beam splitting occurs, the extraordinary ray travels faster than the ordinary ray, resulting in a phase shift between the two components upon exiting the crystal. This can alter the polarisation state of the originally unpolarised light.

On the screen behind the calcite crystal, a splitting of the incident beam into two separate beams is expected. To confirm that these correspond to the ordinary and extraordinary rays, a polariser can be inserted between the calcite and the screen. In one orientation of the polariser, one of the beams disappears, indicating that it is linearly polarised. Rotating the polariser by 90° causes the previously invisible beam to appear, while the other one now vanishes. This confirms that both beams are linearly polarised and that their polarisation directions are orthogonal, as expected for ordinary and extraordinary rays.

Finally, to demonstrate that birefringence disappears when the light propagates along the optical axis, the calcite crystal shold be rotated. When the orientation is such that the incident light travels either parallel or perpendicular to the optical axis, only one beam is visible, even without a polariser between the calcite and the screen. This confirms the disappearance of birefringence under these conditions.

5 Polarisation modification using a quarter-wave plate

This experiment investigates how the polarisation state of light can be modified using retardation plates. A quarter-wave plate made of quartz is used for this purpose. As with calcite (see Section 4), quartz is a birefringent material, meaning that the refractive index — and therefore the speed of light — depends on the direction of propagation relative to the optical axis.

When linearly polarised light enters the quarter-wave plate, it can be decomposed into two components: One component with the electric field parallel to the optical axis (slow axis), one component with the electric field perpendicular to the optical axis (fast axis).

Because these two components propagate at different speeds due to different refractive indices (n_{slow} and n_{fast}), they accumulate a **phase difference** $\Delta\varphi$ as they travel through the plate. This phase difference is given by:

$$\Delta\varphi = \frac{2\pi}{\lambda}\, d\,(n_{\text{slow}} - n_{\text{fast}}) \tag{7}$$

where λ is the wavelength of the light, d is the thickness of the retardation plate, n_{slow} and n_{fast} are the refractive indices for the slow and fast axes, respectively.

A plate is referred to as a quarter-wave plate if the thickness d is chosen such that the resulting phase difference is $\Delta\varphi = \frac{\pi}{2}$

This phase shift can convert linearly polarised light into circularly polarised or elliptically polarised light, depending on the orientation of the incoming polarisation relative to the optical axis of the plate.

5.1 Jones formalism for polarisation states

The Jones formalism can be used to describe mathematically how polarisation states are modified by optical components. A polarisation state is represented by a vector, known as the *Jones vector*. The action of an optical component is described by a matrix, known as the *Jones matrix*. When multiple optical components are placed in the same beam path, their effects are described by the matrix product of their individual Jones matrices.

Linearly polarised in x-direction: $\vec{J}_x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

Linearly polarised in y-direction: $\vec{J}_y = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Linearly polarised at 45°: $\vec{J}_{xy} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Left circularly polarised: $\vec{J}_{\text{circ}-} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$

Right circularly polarised: $\vec{J}_{\text{circ}+} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$

The Jones matrix for a quarter-wave plate with the fast axis aligned along the x-direction is given by:

$$M_{\lambda/4} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1-i & 0 \\ 0 & 1+i \end{pmatrix} = e^{-i\frac{\pi}{4}} \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

This matrix introduces a relative phase shift of $\pi/2$ between the x- and y-components of the electric field, thus converting linear polarisation into elliptical or circular polarisation depending on the input state.

5.2 Polarisation state modification with a quarter-wave plate

In this experiment, linearly polarised light is generated by a polariser. This light then passes through a quarter-wave plate at three different angles: 0°, −45°, and −90°. The angle α is defined as the angle between the direction of polarisation and the optical axis of the quarter-wave plate.

The theoretical results based on Jones formalism are as follows:

- For $\alpha = 0°$, linearly polarised light is preserved with a phase shift:

$$M_{\lambda/4} \cdot \vec{J}_x = ie^{-i\frac{\pi}{4}} \cdot \vec{J}_x$$

- For $\alpha = -90°$, linearly polarised light is again preserved with a phase shift:

$$M_{\lambda/4} \cdot \vec{J}_y = e^{-i\frac{\pi}{4}} \cdot \vec{J}_y$$

- For $\alpha = -45°$, circularly polarised light is obtained:

$$M_{\lambda/4} \cdot \vec{J}_{xy} = e^{-i\frac{\pi}{4}} \cdot \vec{J}_{\text{circ}-}$$

To determine the polarisation state after the quarter-wave plate, a second linear polariser (analyser) is placed behind it.

- For angles 0° and −90°, maximum and minimum intensities are observed depending on the orientation of the analyser, confirming linear polarisation.

- For $\alpha = -45°$, the intensity remains constant regardless of analyser orientation, indicating circular polarisation.

For all other angles between 0° and −90°, elliptically polarised light is produced. In these cases, multiplication of the Jones matrix with a linear input vector results in a vector whose components differ in both magnitude and phase (i.e., one becomes imaginary), corresponding to elliptical polarisation.

5.2.1 Using two quarter-wave plates

At the angle −45°, a second quarter-wave plate is placed directly behind the first with its optical axis aligned parallel to the first. The combined effect of both plates is described by:

$$M_{\lambda/4} \cdot M_{\lambda/4} = -i \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = -iM_{\lambda/2}$$

This corresponds to a half-wave plate, which rotates the polarisation direction of the input vector $\vec{J}_{xy}$ by 90°:

$$M_{\lambda/4} \cdot M_{\lambda/4} \cdot \vec{J}_{xy} = -iM_{\lambda/2} \cdot \vec{J}_{xy} = -i \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Thus, the quarter-wave plate combination acts as a polarisation rotator. This result is confirmed experimentally by observing that the maximum and minimum transmission of light through the second polariser now occur at analyser angles rotated 90° relative to the initial direction.

5.2.2 Material dependence

Replacing the quartz plate with a calcite plate would not fundamentally change the result, as the birefringent behaviour depends on the orientation of the optical axis relative to the incident polarisation, not on the specific material. The critical factor is the anisotropic structure and its cut.

6 Circular birefringence of sucrose

In nature, sucrose is mainly produced from D-glucose and D-fructose, without any L-glucose or L-fructose, leading to an inherent chirality. This chirality causes a rotation of the polarisation direction of light passing through it. We aim to measure this circular birefringence for different sucrose concentrations.

To determine the polarisation direction more precisely, we use Laurent's half-shade polarimeter. It is generally easier for human perception to compare relative brightness levels than to judge a minimum or maximum. Therefore, we use two slightly different polarisation directions that pass through the same polarisation filter and adjust the position until both fields appear equally bright.

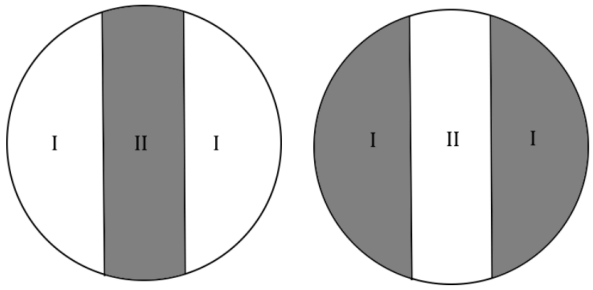

Figure 3: Looking through the polarimeter, there are two fields visible, one at the edges and one in the middle. Field I has a slightly different polarisation angle than field II. The light from both fields passes through an adjustable polarisation filter, causing a difference of brightness.[2]

6.1 Half-shade angle of the polarimeter

The polarisation fields of the polarimeter are aligned as shown in Fig.3. We first determine the half-shade angle of the half-shade polarimeter. This is done by measuring the position of minimal brightness in field II. After five measurements, we obtain an average angle of $\phi_{II} = (170.50 \pm 0.72)°$. For the minimum brightness in field I, the position is measured to be $\phi_I = (182.10 \pm 0.22)°$.

The half-shade angle is defined as the difference between the two measured angles:

$$\Psi = \varphi_{\mathrm{I}} - \varphi_{\mathrm{II}}$$

This yields a half-shade angle of $\theta = (11.60 \pm 0.73)°$.

6.2 Measurement of the sucrose concentration

In this part of the experiment, we send light through a solution of sucrose in water and examine its optical rotation θ. To determine the rotation more precisely, we use the mentioned half-shade polarimeter.

A sucrose-water solution with a concentration of $c = 100\frac{g}{l}$ rotates the light of $\lambda = 595nm$ at an angle $\theta = (14.00 \pm 0.75)°$.The new (thinner) solution rotated the light at an angle $\theta = (9.00 \pm 0.59)°$.

We can now determine sucrose concentrations optically using the following formula:

$$c_{\text{saccharimeter}} = \frac{\theta}{(\theta/c)_1} \tag{8}$$

where θ is the rotation measured with the polarimeter, and $(\theta/c)_1$ is the optical rotation per unit sucrose concentration, determined previously. This allows us to use the polarimeter as a saccharimeter to measure the concentration of an unknown sucrose solution.

With the values from the known solution we calculated a concentration of $c = (64.3 \pm 6.8)\frac{g}{l}$. This seems reasonable for the amount of water we added to the known solution.

7 Measuring the Faraday effect inside a glass body

If a magnetic field is applied to an atom, the Zeeman effect lifts the degeneracy of the quantum states, resulting in different energies proportional to their magnetic moments. This leads to a difference in the refractive indices for σ^+ and σ^- polarised light. Linearly polarised light can be considered a superposition of σ^+ and σ^- components. Due to the differing refractive indices, a phase shift between the σ^+ and σ^- components arises. This phase shift manifests itself as a rotation ϑ of the linearly polarised light.

For an approximately constant magnetic field B along the direction of light propagation, we expect a linear relationship:

$$\vartheta = VBl \tag{9}$$

where l is the path length of the light inside the medium subjected to the magnetic field, and V is the Verdet constant.

In this part of the experiment, we aim to determine the Verdet constant for a sample made of NSF6 high-homogeneity glass. The magnetic field is generated by a pair of solenoids positioned in front of and behind the glass sample. Since the magnetic field cannot be directly measured while the glass body is in place, we first measure the magnetic flux density as a function of the solenoid current. Furthermore, because the magnetic field is not uniform, we must measure the field distribution along the light path and determine the relationship between the effective magnetic field B_{eff} and the maximum magnetic field $B_{\max}$.

7.1 Effective magnetic flux density

To obtain the relationship between the effective magnetic field strength and the maximum field strength, we measure the magnetic flux density at various positions between the two coils using a Hall probe. By numerically integrating the measured data points and dividing by the total length, we calculate the average (effective) magnetic field for $I = 1.49A$ (see Fig. 4) as

$$B_{\text{eff}} = (63.34 \pm 0.35)\,\text{mT}.$$

The current in the solenoid was The ratio $\alpha = \frac{B_{\text{eff}}}{B_{\text{max, measured}}}$ can then be computed using the measured maximum magnetic flux density $B_{\text{max, measured}} = (98.0 \pm 2.0)\,\text{mT}$, yielding $\alpha = 0.646 \pm 0.013$

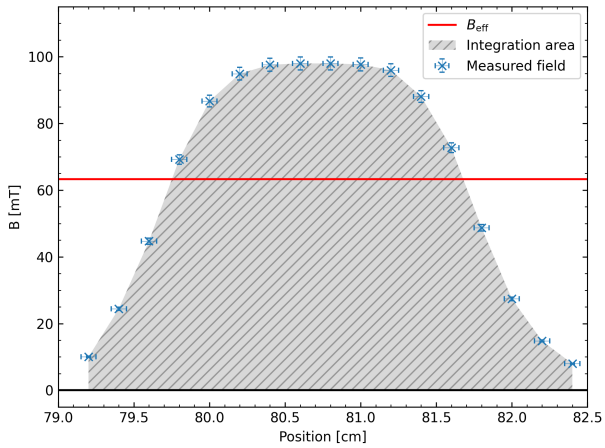

Figure 4: Magnetic flux densities at different positions. To take the average B_{eff} of the magnetic field, we numerically integrate over the measured field and divide by the position range.

We assume that α remains constant for different currents, as the magnetic field geometry should not change with the glass sample in place.

7.2 Magnetic flux density per current

Now the relationship between the magnetic flux density and the current generating the magnetic field is investigated. For this purpose, the maximum magnetic field values $B_{\max}$ are measured using a Hall probe for various currents I. According to Ampère's law, the magnetic field is expected to increase proportionally with the current.

As shown in Fig.5, $B_{\max}$ exhibits an approximately linear dependence on I. Therefore, a linear regression is performed, resulting in the slope

$$C_B = \frac{\mathrm{d}B_{\max}}{\mathrm{d}I} = (61.076 \pm 0.025)\,\frac{\text{mT}}{\text{A}}.$$

7.3 Rotation per current

The glass body is placed between the two solenoids. A filter is used in front of a lamp that only allows light with a wavelength between 400 nm and 560 nm to pass through. The light is directed through the interference filter and a polariser, then travels through the glass body, passes through another polariser (the analyser), and finally reaches a sensitive photodiode.

The orientation of the analyser is adjusted until the photodiode detects a minimum current. This position corresponds to the rotation angle ϑ of the polarisation of the light caused by its passage through the glass body. The rotation is measured for solenoid currents ranging from -3 A to 3 A.

A linear regression (see Fig.6) yields the following rotation angle per unit solenoid current:

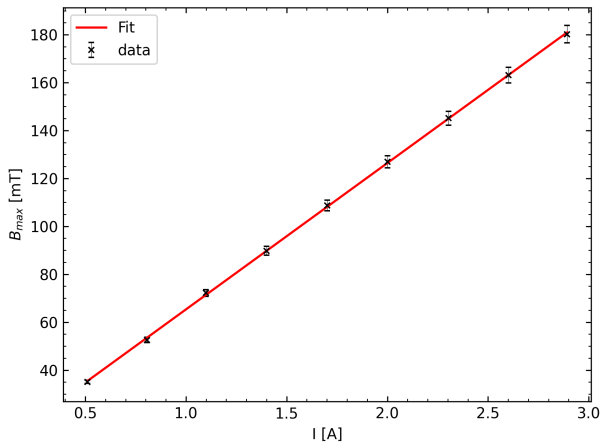

Figure 5: Maximal flux densities for different currents. The proportionality constant between the strongest magnetic flux density and the solenoid current is determined by conducting a linear regression.

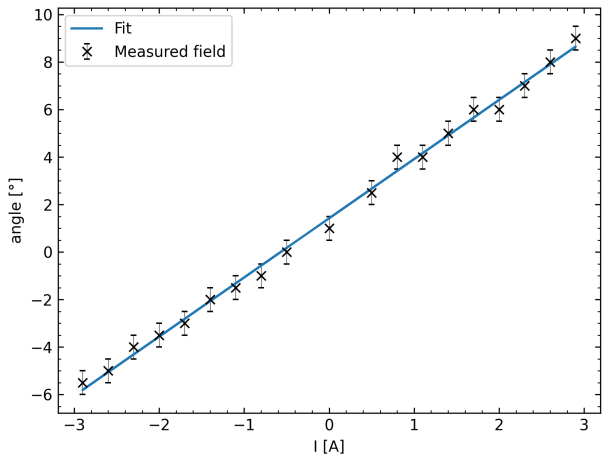

Figure 6: Rotation per current for 400-560nm. A linear regression is used to determine the proportionality between solenoid current and the rotation angle θ.

wavelength in nm	slope in $\frac{°}{A}$
400-560	2.492 ± 0.038
560-630	1.796 ± 0.036
480-570	2.354 ± 0.040
360-460	2.122 ± 0.054

Table 1: We measured the rotation dependent on the current for different wave lengths and calculated the slope.

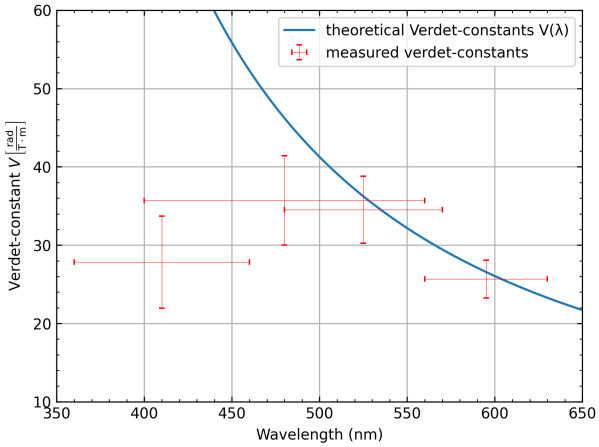

Figure 7: Verdet constants of NSF6 as a function of the wavelength. The experimental values are shown together with the theoretical trend, obtained with the Sellmeier formula. Due to a defect in the interference filter in the range of 360–460 nm, hardly any light was transmitted in this spectral region, making the measurement in this range unreliable.

$$m_\vartheta = (2.492 \pm 0.038)°/\mathrm{A}.$$

Since the solenoids heat up at higher currents, we suspect that the magnetic fields in some measurements may be weakened due to Curie's law [2]. This would cause the actual value of m_ϑ to be larger than the measured value.

7.4 Calculating the Verdet constant

We can calculate the Verdet constant from the proportionality constants we determined previously using the following formula:

$$V = \frac{\vartheta}{lB_{\text{eff}}} = \frac{2\pi}{360°} \cdot \frac{m_\vartheta}{\alpha C_B l}$$

Using the angle in radians, we obtain the Verdet constants shown in Tab. 2. Alternatively, one can calculate the Verdet constant theoretically using the following expression:

$$V = \frac{e}{2m_e c} \cdot \lambda \cdot \frac{dn}{d\lambda}$$

wavelength	Verdet constant $[\frac{rad}{T*m}]$
400-560	35.7 ± 5.7
560-630	25.7 ± 2.4
480-570	34.5 ± 4.3
360-460	27.8 ± 5.9

Table 2: Verdet constants for different wave lengths.

where e is the elementary charge, m_e the electron mass, c the speed of light, λ the wavelength, and n the refractive index. The experimental value agrees with the theoretical expectation within the margin of error, but not for the first point. Due to a defect in the interference filter in the range of 360–460 nm, hardly any light was transmitted in this spectral region, making the measurement in this range unreliable.

8 Conlusion

In the first experiment, light from a lightbulb is sent through a polarisation filter to obtain linearly polarised light. The light then travels through a second adjustable polarisation filter into a photocell to measure the intensity as a function of the angle between the polarisation directions of the polarisation filters. Due to a mistake we can not conclude that Malus' law describes the intensity for different filter angles.

In addition, it was examined how a quarter-wave plate affects linearly and circularly polarised light. A quarter-wave plate turns linear polarisation into circular polarisation if the angle α between the optical axis and the incoming polarised light is 45°. It can also turn circular polarisation back into linear polarisation by placing a second quarter-wave plate behind the first one. For the angles of 0° and −90°, the light remains linearly polarised.

The half-shade angle of a half-shade polarimeter was determined. Then, the half-shade polarimeter was used to measure how the polarisation direction of linearly polarised light is rotated by chiral sucrose. The largest rotation was observed for a wavelength of 595 nm . This was used to determine the concentration of sugar in a different solution.

Finally, the Faraday effect was recreated. Linearly polarised light of different wavelengths were sent through N-SF6 high-homogeneity glass, which was exposed to a magnetic field. The rotation for different magnetic field strengths was measured. A linear dependence between the magnetic field and the polarisation rotation was observed and the Verdet constants in N-SF6 glass were calculated. This experimental value agrees with the theoretical trend of the N-SF6 glass obtained from the Sellmeier formula.

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9 Appendix

Theoretical Value of $\frac{dn}{d\lambda}$

Consider the Sellmeier formula:

$$n^2 - 1 = \frac{B_1\lambda^2}{\lambda^2 - C_1} + \frac{B_2\lambda^2}{\lambda^2 - C_2} + \frac{B_3\lambda^2}{\lambda^2 - C_3} \tag{10}$$

with the constants B_1, B_2, B_3, C_1, C_2, and C_3 taken from [3].

By differentiating Eq. 10 with respect to λ, the following expression is obtained:

$$2n\frac{dn}{d\lambda} = -\frac{2B_1C_1\lambda}{(\lambda^2 - C_1)^2} - \frac{2B_2C_2\lambda}{(\lambda^2 - C_2)^2} - \frac{2B_3C_3\lambda}{(\lambda^2 - C_3)^2} \tag{11}$$

Using the Sellmeier formula again to express n in terms of λ, we get the more explicit formula for the derivative:

$$\frac{dn}{d\lambda} = \left(-\frac{B_1C_1\lambda}{(\lambda^2 - C_1)^2} - \frac{B_2C_2\lambda}{(\lambda^2 - C_2)^2} - \frac{B_3C_3\lambda}{(\lambda^2 - C_3)^2}\right)\left(\frac{B_1\lambda^2}{\lambda^2 - C_1} + \frac{}{\lambda} \right. \tag{12}$$

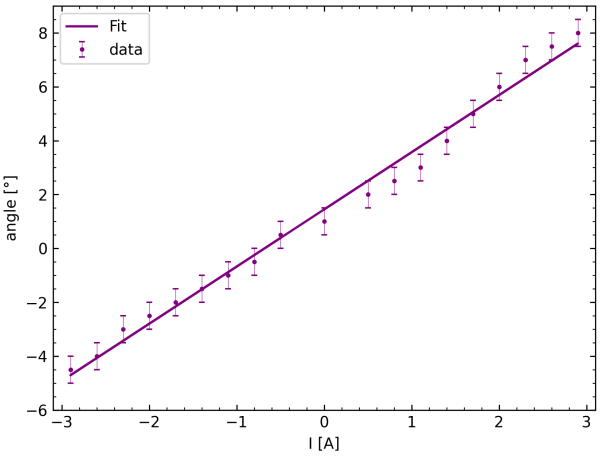

Figure 8: Rotation per current all nm